Building and evaluating a cross-hazard machine-learning approach to unified lunar forecasting

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Abstract

LIPAS (Lunar Impact and Predictive Analysis System) is a hybrid forecasting and decision-support platform that combines machine learning with physics-based computational models to predict the Moon's space-weather conditions. These conditions include temperature, radiation, dust activity, micrometeorite impacts, solar storms, and moonquakes. By unifying space-weather data products from NASA and the National Oceanic and Atmospheric Administration (NOAA) alongside OMNI2 data, the LIPAS platform processes the data through a temporal Long Short-Term Memory (LSTM) neural network. Users can view the LIPAS space-weather predictions via an interactive, web-based dashboard for any lunar location.

This study addresses three distinct research questions. First, it documents how the hybrid model LIPAS was developed and what was learned throughout the process. Second, it studies how operational usability can be improved in real mission scenarios compared with existing prediction systems. Third, this research investigates whether integrating machine learning and physics-based modeling into LIPAS improves prediction accuracy compared with existing single-purpose NASA and NOAA lunar environmental models.

This work highlights how combining scientific research, advanced software engineering, and diverse datasets creates a practical tool that could be used to support safer and more predictable lunar exploration in the future. Because simultaneous in-situ observations are unavailable for all six hazards, evaluation uses archived measurements, observation-constrained estimates, engineering-model output, and physics/catalog proxies rather than treating every target as direct ground truth. Two 14-day windows at three sites show lower solar- ersus selected replicas in both windows (Window B radiation reported with a quiet-window caveat: hybrid RMSE 0.0188 versus replica 0.00266 temperature, moonquakes, and dust stay physics-aligned; micrometeorite contrast is limited to the shower te

Keywords: lunar environment, space weather, hybrid forecasting, machine learning, long short-term memory, mission decision support

1. Introduction

Renewed international interest in lunar exploration, particularly through NASA's Artemis program and its international partners, has increased the need for reliable and accessible information about the lunar environment (NASA, 2020). The Moon has no protective atmosphere or global magnetic field, leaving its surface exposed to large temperature variations, radiation, micrometeorite impacts, electrostatic dust, and seismic activity. These hazards can affect crew safety and the operation of spacecraft and surface equipment.

A range of scientific models already exists for studying individual lunar hazards. These models, however, are generally designed for specific hazards and operate as separate tools. Their outputs are difficult to combine into a single operational view or use together for forecasting. Mission planners may therefore need to consult several data sources and models when assessing environmental conditions, which complicates decision making.

The Lunar Impact and Predictive Analysis System (LIPAS) was developed to address this fragmentation by bringing multiple lunar-hazard forecasts into a single decision-support platform. The system combines machine-learning predictions with physics-based models, public heliophysics data[1], and adaptive weighting between the two approaches. The literature reviewed here describes specialized lunar hazard models and archives, while LIPAS explores how these sources can be combined with machine learning in a unified decision-support framework.

This study addresses three research questions:

RQ1:
Can a hybrid lunar-forecasting system be developed by effectively combining machine-learning and physics-based approaches?

RQ2:
Can integrating multiple lunar environmental hazards and data sources into a unified forecasting and decision-support system improve operational usability?

RQ3:
Does the locked LIPAS hybrid improve prediction performance for selected lunar hazards relative to selected replicated public baselines and physics-only references?

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[1] Heliophysics studies the Sun, the solar wind, and their interaction with planetary environments throughout the heliosphere.

LIPAS is not intended to replace established scientific models or flight-qualified agency products, but to combine public data and reproducible model paths within a common forecasting framework. The main contribution of this work is the development and evaluation of a unified hybrid system that retains physics-based constraints while using machine learning where it can provide additional predictive information. In addition to forecasting multiple hazards within one platform, the system is intended to make these forecasts easier to interpret and use in mission-planning workflows.

2. Overview of current models

2.1. General overview

The study of lunar environmental conditions has historically relied on single-purpose models, each focused on a particular hazard. Although these models incorporate advanced scientific understanding in their respective domains, they remain limited when applied to operational mission planning because they do not provide integrated insights or unified predictive support. Each model was built to solve a specific scientific problem rather than to operate as part of a broader forecasting system. This creates a gap because the interactions among hazards cannot readily be analyzed.

2.2. NASA DONKI model

The NASA Database of Notifications, Knowledge, Information (DONKI) API provides near-real-time monitoring of solar flares, coronal mass ejections (CMEs), and solar energetic particle (SEP) events (NASA, 2023b). DONKI is valuable as a space-weather event catalog for radiation context, but it is event-driven and primarily reactive, with limited temporal coverage and incomplete data for continuous mission planning. It does not itself produce multi-day forecasts or integrate other hazard types, making it insufficient as a stand-alone planning tool.

2.3. Temperature models

Temperature predictions rely heavily on the Diviner Lunar Radiometer Experiment (Paige et al., 2010; Williams et al., 2017; NASA, 2023a), which provides detailed infrared measurements of surface temperature across the Moon. These measurements accurately characterize lunar thermal conditions, including uneven terrain, but do not themselves produce short-term forecasts. Apollo-era heat-flow measurements are similarly sparse and lack the temporal resolution required for short-term forecasting (Langseth et al., 1972). Neither source was designed to predict future temperature; both principally describe measured or reconstructed conditions.

The early prototype used a simplified Diviner-style dayside surface-temperature relation,

$$T = 392 \cos^{0.25}(z) - 273, \tag{1}$$

where T is surface temperature in degrees Celsius and z is the solar zenith angle at the target coordinates. Legacy nightside profiles use polynomial fits derived from Apollo heat-flow measurements. Although deterministic, these stand-alone relations do not dynamically adjust to changing space-weather conditions.

2.4. Lunar dust models (LADEE/LDEX)

The lunar dust environment has been studied by the Lunar Atmosphere and Dust Environment Explorer's Lunar Dust Experiment (LADEE/LDEX), which documented dust-particle density and electrostatic activity (Horanyi et al., 2015; NASA Planetary Data System, 2015). LDEX is important for understanding surface hazards, but its data are limited to the LADEE mission epoch and orbital coverage, providing neither continuous global monitoring nor direct predictive capability. The mission ended in 2014, leaving substantial temporal and spatial gaps.

2.5. Moonquakes and Apollo PSE

Moonquake monitoring relies on the Apollo Passive Seismic Experiment (PSE) catalog. Recent reanalysis identified thousands of previously undetected moonquakes (Onodera, 2024). The PSE dataset is retrospective, cannot provide real-time forecasting, and does not integrate other hazards. Nevertheless, it contains statistical patterns tied to the lunar tidal cycle. LIPAS uses these patterns as physics/catalog context for moonquake-rate estimates. Because deep moonquakes occur in clusters correlated with tidal stress between Earth and the Moon, recurring catalog patterns can be projected forward statistically even without real-time seismic sensors (Kawamura et al., 2017; Lammlein, 1977; Nakamura, 2005; Nunn et al., 2020).

2.6. Micrometeorite prediction with MEM 3

NASA's Meteoroid Engineering Model (MEM 3) estimates meteoroid collision rates (Moorhead et al., 2020). It is precise within its engineering domain but does not directly incorporate secondary operational effects such as dust lofting or the interactions of meteor-shower timing with surface operations. LIPAS therefore applies shower-calendar modulation and dust coupling to the MEM 3 baseline used in its physics path (American Meteor Society, 2026; International Meteor Organization, 2026; Szalay and Horanyi, 2015).

2.7. GOES-18 radiation context

NOAA's Geostationary Operational Environmental Satellite 18 (GOES-18) continuously monitors solar X-ray flux and energetic-particle counts (National Oceanic and Atmospheric Administration, 2023). GOES measurements are made in Earth orbit and must be adapted for lunar-surface conditions because the Moon lacks a global magnetosphere. GOES-18 does not provide a dedicated multi-day lunar radiation forecast and was not designed specifically for lunar missions.

2.8. Summary

Individual models and archives such as DONKI, Diviner, MEM 3, PSE, and LDEX are essential for scientific research, but their limitations in coverage, integration, and predictive usability motivate a unified system. LIPAS combines the strengths of these single-purpose approaches with real-time NASA and NOAA space-weather data, historical lunar measurements, and a hybrid machine-learning architecture. The resulting framework is proactive and integrative, providing a comprehensive interface for mission planning and safety assessment while retaining the limitations of its source data and models.

3. System overview: LIPAS

LIPAS is a hybrid forecasting system that combines machine learning with physics-based calculations. Rather than using machine learning for every hazard, LIPAS dynamically negotiates trust between neural predictions and physics-based equations. Established physical relationships remain the main basis for hazards that are already well described by physics, while machine learning provides corrections where relationships among changing environmental variables may contain useful predictive information. This architecture is intended to be scalable and risk-aware for future crewed and uncrewed missions.

3.1. Prototype limitations and migration to a backend

The original LIPAS prototype was a browser-based application in which data processing and machine learning were performed on the client side. As it grew, forecasts for physically constrained hazards such as temperature varied between training runs. This showed the need to retain stronger physical relationships and move processing to a backend that could support a more stable data and forecasting service.

The application was therefore re-engineered as a server-assisted hybrid model with a Python backend. Data filtering, API polling, model training, and inference are handled on the server, which supplies a stable data service to the front-end dashboard. The current system can assign different levels of influence to machine learning and physics according to the hazard and current conditions.

3.2. Hybrid forecasting and uncertainty architecture

The core forecasting model uses an 18-feature input representation that contains hazard-related, temporal, spatial, and space-weather variables. Rather than processing only current conditions, the model receives a 72-step temporal window. It produces six target outputs: solar activity, radiation, temperature, moonquake rate, micrometeorite flux, and dust density. The larger input representation allows information from one environmental channel to contribute to the prediction of another. The temporal trunk is based on a long short-term memory architecture (Hochreiter and Schmidhuber, 1997).

A physics-estimation function first calculates a physically grounded value for each hazard. A subsequent refinement step combines that value with the neural-network output, allowing machine learning to provide a correction instead of learning the complete physical relationship. For hazard channel i, the resulting prediction is

$$\widehat{Y}_{\text{hybrid},i} = w_i \widehat{Y}_{\text{ML},i} + (1 - w_i)\widehat{Y}_{\text{physics},i}. \tag{2}$$

The weight is evaluated from the current space-weather regime and the neural network's uncertainty:

$$\begin{aligned} w_i = \max\{0, \min[1,\ & (w_{\text{base},i} + \Delta_{\text{regime},i}) \\ & \times \max(0.35, 1 - 1.6\frac{\sigma_{\text{ML},i}}{S_i})]\}. \end{aligned} \tag{3}$$

Here $w_{\text{base},i}$ is the baseline machine-learning weight, $\Delta_{\text{regime},i}$ is an adjustment for the current storm posture, $\sigma_{\text{ML},i}$ is the neural uncertainty estimate, and S_i is the physical operating span of the channel. Uncertainty is estimated with Monte Carlo dropout: the network is evaluated repeatedly while a randomly selected subset of neural connections is disabled on each pass. Closely grouped outputs indicate a stable learned prediction, whereas a wider spread indicates greater uncertainty in the learned model.[2] When $\sigma_{\text{ML},i}$ rises, Eq. (3) reduces the neural contribution and shifts the forecast toward the physics baseline.

Where physics provides a strong basis, LIPAS assigns greater weight to the physics path. In the locked production blend, temperature, moonquakes, micrometeorite flux, and dust are predominantly physics-led, while solar activity and radiation retain larger machine-learning weights. The backend also supports optional ensemble forecasts for uncertainty estimation, although the evaluated production path uses a single locked temporal model.

3.3. Development of the hybrid framework

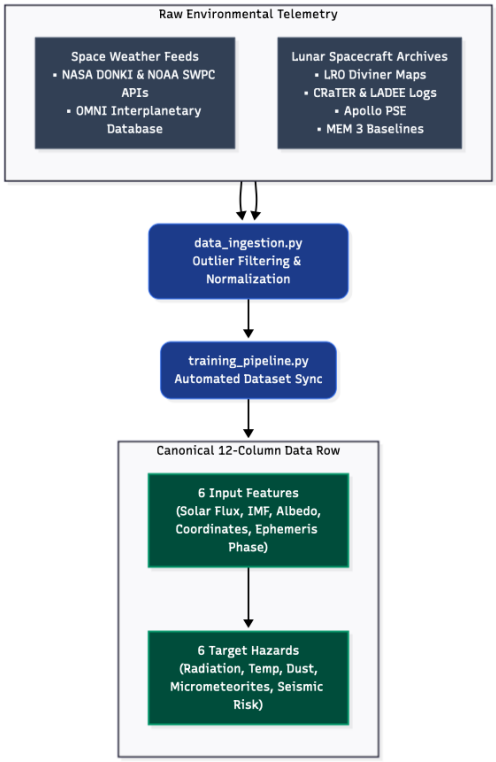
The current LIPAS implementation uses HTML, CSS, and JavaScript for the front end and a Python/TensorFlow backend for data ingestion, model training, physics blending, and forecast serving (Google TensorFlow Team, 2026; Chollet et al., 2015). Multiple production training runs were completed. The reported quantitative evaluation used the second training-run model because it was the best-performing completed production model when the model was locked for testing. Its chronological held-out test RMSE was approximately 4.97, compared with approximately 13.66 for the corresponding persistence baseline; no later completed run had surpassed it at the evaluation cutoff.

Earlier prototype experiments used the browser-based neural network that preceded the Python hybrid backend. They are treated as developmental diagnostics rather than evidence of current model performance because they lacked the architecture and baseline framework used in the reported evaluation. The locked production model was subsequently evaluated using the framework in Section 4.

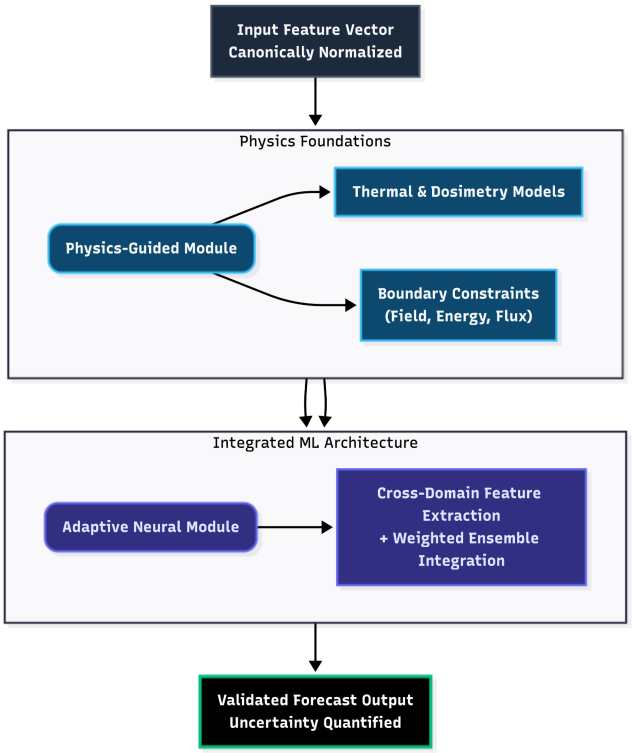
3.4. Data ingestion and pipeline orchestration

The backend handles real-time data streams through a modular ingestion pipeline that queries, cleans, and normalizes external APIs into the 18-feature representation. The pipeline joins CME events from DONKI, solar indices from the NOAA Space Weather Prediction Center, and interplanetary magnetic-field data from OMNI (NASA, 2023b; National Oceanic and Atmospheric Administration, 2023; NASA Goddard Space Flight Center, 2026; King and Papitashvili, 2005). These space-weather feeds are combined with LRO Diviner thermal maps, CRaTER radiation logs, and LADEE/LDEX dust measurements (NASA, 2023a,c; NASA Planetary Data System, 2015). Long-term baselines include Apollo PSE data, Apollo heat-flow measurements, MEM 3, and peer-reviewed micrometeorite-flux data.

[2] This is epistemic uncertainty: uncertainty caused by limitations in the model or its training-data coverage.



(a) Hybrid forecast core: physics establishes the channel baseline, the temporal model supplies a correction, and uncertainty controls the blend.



(b) Data path from agency feeds and lunar archives through normalization, inference, validation, and dashboard delivery.

Figure 1: End-to-end LIPAS architecture. The two flow diagrams show how source data reach the uncertainty-aware hybrid forecast and operational interface.

The end-to-end process coordinates data collection, model training, and a two-standard-deviation outlier filter. Values outside that threshold are flagged as potential anomalies and prevented from entering the training stream.

3.5. Command-center dashboard and map visualization

The user interface was overhauled after the prototype limitations were identified. The resulting mission-control-style interface uses CSS Grid and Flexbox and consists of a dashboard, mission planner, and Cesium globe (Cesium GS, 2026; Chart.js Contributors, 2026). The dashboard includes more than 50 named lunar locations and a coordinate search interface for arbitrary sites. It provides topographic maps with hazard overlays, 30-day outlooks, particle-flux visualizations, and warnings and recommendations for mission planning. Historical and predicted trends can be displayed as line, bar, or area charts.

The mission-planning interface accepts mission parameters and duration targets. Based on available forecasts, it proposes EV A departure windows, assesses landing-zone conditions, and generates an exportable mission plan. A Luna Assistant provides additional hazard information. The Cesium page supplies an interactive two- or three-dimensional lunar map with time-dependent overlays.

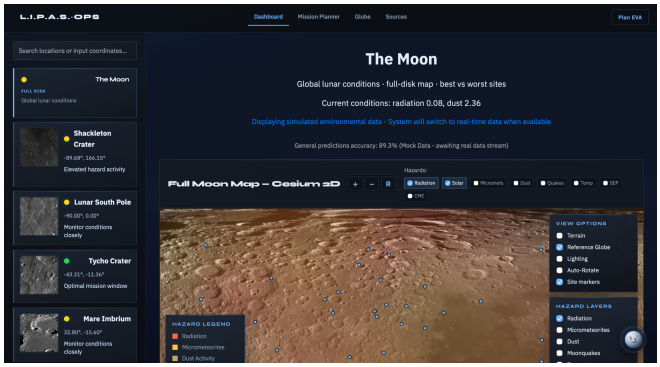
4. Methodology and experimental design

4.1. Experimental design and evaluation framework

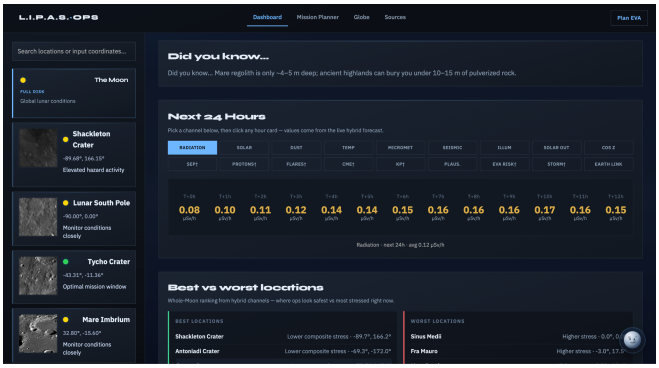
The research was divided into three phases, each addressing one research question. The first focused on developing LIPAS and identifying lessons from that process. The second examined operational use in a simulated mission. The third evaluated whether the locked hybrid system improved predictive performance relative to established physics-based and agency-style reference approaches. All reported quantitative evaluations used the locked production model from the second training run.

4.1.1. Phase 1: development of the hybrid model

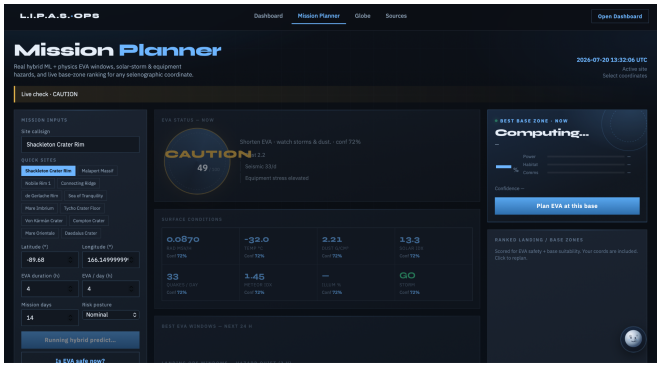
The first phase examined the transition from the original browser prototype to the hybrid system. It included an early 14-day daytime prototype evaluation at 0°N, 0°E from April 5 to April 19, 2026. The purpose was not to establish operational accuracy, but to identify weaknesses in the client-side machine-learning approach and determine which channels required a physics-based foundation. Treating all six hazards as machine-learning problems produced unstable results for strongly physics-constrained hazards. Consequently, temperature, moonquakes, dust, and micrometeorites were kept predominantly physics-based, while machine learning was given a larger role in solar activity and radiation.



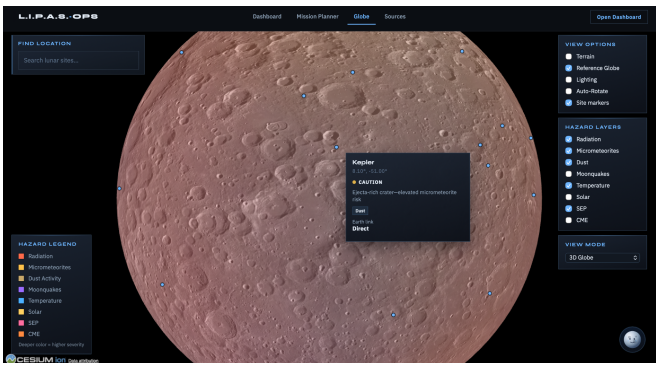
(a) Site selection, coordinate search, and topographic hazard map.



(b) Forecast drivers, status, and diagnostics.



(c) Exportable mission report and risk summary.



(d) Interactive lunar globe and hazard overlays.

Figure 2: Representative LIPAS operator workflow from site selection and diagnostic review to mission reporting and global visualization. Additional interface views are provided in the supplementary material. The public prototype is available at `https://lipas.netlify.app`.

This phase also established the 18-feature input structure, the 72-step temporal window, physics-estimation paths, and adaptive hybrid weighting. These components were combined into the locked model used in the remaining phases.

4.1.2. Phase 2: operational mission evaluation

The second phase examined whether integrating several environmental hazards could support operational usability during a simulated lunar mission. A controlled radiation-focused EV A scenario used a timeline of 7 h 37 min, matching the duration of Apollo 17 EV A. LIPAS was monitored at fixed intervals and compared with the information an operator would receive from separate public NASA and NOAA products, including GOES, NOAA SWPC, and DONKI. The same predefined caution and no-go thresholds were applied to each pathway. The comparison focused on when each system produced warnings and when the simulated EV A reached caution or no-go conditions. It did not establish that LIPAS is safer than NASA systems or measure usability with mission personnel.

4.1.3. Phase 3: quantitative prediction evaluation

The third phase evaluated the current locked hybrid system against physics-based and replicated NASA/NOAA reference approaches. Two 14-day windows were used at three locations: an equatorial reference at 0°N, 0°E; Apollo 11; and the Nobile Rim Artemis III candidate region (Fig. 3). Window A covered July 5 to July 18, 2026, and was predominantly lunar night; Window B covered July 22 to August 4, 2026, and was predominantly lunar day. For every site and date, LIPAS predicted solar activity, radiation, temperature, moonquake rate, micrometeorite flux, and dust density.

Predictions were compared with persistence, physics-only references generated from the same `physics_estimate` path, and replicated public baselines described in Section 4.4. Archived OMNI, NOAA, GOES, and DONKI data allowed the experiments to be replayed under common dates, locations, input conditions, and targets rather than depending on changing live data.

Continuous co-located measurements are unavailable for all six hazards across the lunar surface. Consequently, the term *evaluation label*, rather than *ground truth*, is used for the comparison target unless a direct observation exists. Each channel was classified before scoring as an archived measurement or driver, an observation-constrained estimate, an engineering-model output, or a physics/catalog proxy (Table 1). No aggregate result in this paper represents validation against simultaneous in-situ measurements of all six hazards.

Performance was evaluated primarily by root mean square

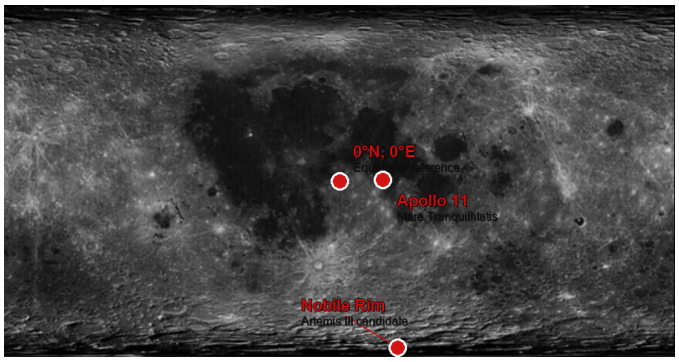


Figure 3: Lunar locations used in the reported studies: the equatorial reference, Apollo 11, and Nobile Rim.

error (RMSE) and mean absolute error (MAE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(\widehat{y}_j - y_j)^2}, \tag{4}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n}|\widehat{y}_j - y_j|. \tag{5}$$

The Window B micrometeorite comparison used the MEM 3 seasonal background with American Meteor Society (AMS) shower modulation for LIPAS, while the replicated comparison omitted that additional shower term. This tests the effect of explicitly representing shower timing during the Perseid period, not general superiority over NASA's complete meteoroid-modeling framework.

An ablation analysis removed selected inputs while holding the remaining conditions fixed. Conditions included the full hybrid model, removal of solar/storm drivers for radiation, removal of micrometeorite context for dust, an ML-only path, and a physics-only path. Resulting predictions were compared with the full system using RMSE and MAE.

4.2. Qualitative-comparison criteria

The operational comparison used nine ordinal criteria. The categories were not converted into numerical scores or combined into an overall rank. The following rules were fixed before the systems were classified:

Hazard coverage.
None: no hazard addressed; Very Low: one hazard; Low: up to two hazards; Moderate: three to four hazards; High: five hazards; Very High: six or more integrated hazards.

Data-fusion capability.
None: one dataset; Low: multiple datasets without integration; Moderate: partial integration; High: integrated multi-source system; Very High: real-time dynamic multi-source fusion.

Temporal forecasting ability.
None: archival only; Low: event-driven alerts without a time horizon; Moderate: short-range estimates for a narrow domain; High: multi-day forecasting for one or more hazards; Very High: multi-day forecasting across all hazards.

Spatial coverage.
None: no spatial resolution; Very Low: specific historical sites; Low: one region or polar area; Moderate: broad regional coverage; High: global or near-global coverage; Very High: complete lunar-surface coverage with location selection.

Real-time operational capability.
None: historical only; Low: archival access with significant delay; Moderate: near-real-time with limited updates; High: live feeds with regular updates; Very High: continuous real-time data with automated processing.

Robustness and stability.
Low: limited coverage or known gaps; Moderate: functional with notable external dependencies; High: well-validated and stable in a narrow domain; Very High: continuously calibrated mission-grade instrumentation with a long operational history.

Validation strength.
None: no comparison; Low: informal or limited comparison; Moderate: structured comparison over a limited period; High: peer-reviewed validation against direct measurements; Very High: direct physical measurement, self-validating by design.

Usability and operational readiness.
Very Low: raw scientific data requiring specialist processing; Low: technical interfaces; Moderate: accessible API or web portal; High: clear operational interface; Very High: interactive dashboard for non-specialist mission planners.

Openness and accessibility.
Low: restricted or proprietary; Moderate: archives requiring registration or processing; High: open API or published datasets; Very High: fully open browser-based access without setup.

4.3. Baseline configurations and control environments

Specific baseline rules were established across all environments. Reference predictions were obtained from public databases, peer-reviewed formulas, engineering-model exports, and agency documentation. In this paper, a replicated NASA/NOAA reference means an author-implemented reconstruction from those public materials; it does not mean that internal, proprietary, operational, or flight-certified agency software was executed. MEM 3 is treated separately because the comparison used an engineering-model export available to the project. Where an established product did not provide a direct multi-day lunar prediction for a study interval, the relevant public method was reproduced as closely as the documented equations and open data allowed.

For every hazard, the study recorded the source, procedure, and whether the target was a direct measurement, engineering baseline, or physics proxy. In the mission simulation and ablation tests, coordinates and physical constants were held fixed while selected inputs were altered. The mission scenario used archived quiet conditions as a background and injected controlled changes, with status checked at fixed intervals. The ablation tests replaced selected driver or hazard inputs with their Window A means as sensitivity tests.

Table 1: Provenance and interpretation of the six quantitative evaluation targets.

Channel	Principal target construction	What is directly observed	Validation status
Solar activity	Archived OMNI/SWPC indices and DONKI event context mapped to the study's solar-activity label	Solar-wind, field, flare, and geomagnetic drivers	Constructed label from measured drivers
Radiation	CRaTER-calibrated background combined with archived solar and proton-event terms	Orbital particle and space-weather measurements	Observation-constrained lunar-surface proxy
Temperature	Diviner/Apollo-constrained thermal equations evaluated at each site and time	Orbital temperature maps and historical heat-flow measurements	Observation-constrained physics proxy
Moonquakes	Apollo PSE catalog rates with tidal, terminator, and location adjustments	Historical Apollo seismic events	Historical-catalog proxy
Micrometeorites	MEM 3 engineering output with a separately stated AMS shower-timing term	MEM 3 output and terrestrial shower timing	Engineering-model/shower proxy
Dust	LDEX-informed background with impact, local-time, and terminator terms	Historical orbital dust detections	Observation-informed physics proxy

4.4. Baseline replication and model specifications

4.4.1. Temperature

Temperature baselines used lunar thermal physics constrained by Diviner observations and Apollo nightside measurements (Vasavada et al., 1999; Paige et al., 2010; Hayne et al., 2017; Williams et al., 2017; Langseth et al., 1972). On the dayside, surface temperature was computed from the Stefan-Boltzmann balance

$$T = \left(\frac{(1-\alpha) I_0 \cos\theta}{\epsilon \sigma} \right)^{1/4}, \tag{6}$$

where I_0 is solar irradiance adjusted for Earth-Sun distance, α is surface albedo[3], ϵ is emissivity, σ is the Stefan-Boltzmann constant, and θ is the local solar-incidence angle. Nightside temperatures were calculated with an Apollo-based cooling curve and adjusted for latitude and a small interior heat term. Site reflectivity, emissivity, terrain, and thermal inertia were taken from the LIPAS site data. Where available, Diviner observations could further constrain the estimate. The early prototype also used Eq. (1); the reported studies used Eq. (6) and the Apollo-based nightside model.

An equivalent form used in the implementation is

$$(1-\alpha) I_s = \epsilon \sigma T^4. \tag{7}$$

4.4.2. Solar activity and radiation

Solar and radiation baselines were built from public space-weather data rather than a direct lunar radiation forecast. Solar activity used OMNI2 and SWPC quantities including Kp, the interplanetary magnetic-field component B_z[4], solar-wind speed and density, and F10.7 where available. DONKI flare, CME, geomagnetic-storm, and SEP records were included for the study dates. The radiation path used a CRaTER-based galactic-cosmic-ray background of approximately 0.38 mSv day^{-1}, converted to an hourly value and adjusted for solar activity (Spence et al., 2010; Schwadron et al., 2014; NASA, 2023c). Additional quiet, storm, and SEP terms used Kp, southward B_z, high solar-wind speed, GOES proton measurements, and solar activity. A small latitude adjustment was applied near the lunar poles. GOES X-ray and proton measurements were model inputs, not direct lunar-surface radiation measurements.

4.4.3. Micrometeorites

Micrometeorite baselines were replicated from the project's MEM 3 export (Moorhead et al., 2020). The annual flux set the principal background level, and available high- and low-density ranges represented seasonal variation. LIPAS added AMS-based meteor-shower modulation separately from the basic MEM 3 baseline. A latitude term was applied only in the LIPAS physics path. For Window B, the replicated comparison used MEM 3 without the AMS shower term, while LIPAS used MEM 3 plus shower modulation. This directly isolates the effect of meteor-shower timing.

4.4.4. Dust

Dust baselines used LADEE/LDEX data rather than a live global forecast. The model combined a normal background with dust associated with micrometeorite impacts, enhanced dust near

[3]Albedo is the fraction of incoming radiation reflected by a surface; emissivity describes how efficiently that surface emits thermal radiation.

[4]The interplanetary magnetic field is carried outward by the solar wind; a sustained southward B_z component can indicate stronger coupling with Earth's magnetic environment.

terminators[5], an optional LDEX local-time factor, and a small regolith-depth effect[6] (Colwell et al., 2007; Horanyi et al., 2015; Stubbs et al., 2006). It is therefore a physics proxy based on LDEX observations, not a claim of continuous dust measurement at the Artemis sites.

4.4.5. Moonquakes

Moonquake baselines used the Apollo PSE catalog. The model combined deep, shallow, and thermal moonquake rates, enhanced activity near terminators, a small equatorial preference for deep events, and a weak regolith effect. No real-time lunar seismometer network was assumed. The output is a catalog-based physics proxy rather than a direct measurement of current activity at the study sites.

Persistence baselines assumed that the latest known value remained constant. Physics-only references were generated from the common replay data. RMSE and MAE were calculated for LIPAS against persistence, physics-only paths, and each replicated reference baseline. The mission simulation used common LIPAS outputs and baselines throughout the Apollo 17-length timeline, and the ablation study used the same replay data while removing selected inputs or pathways.

4.5. Use of AI-assisted tools in software development

AI-assisted tools were used in a supporting role during software development. Cursor and Google Gemini assisted with selected code suggestions, debugging, technical explanations, and user-interface development. All suggested code was reviewed, edited, and tested by the author before use. These tools did not replace the documented physical formulations, create experimental observations, or independently determine the reported results. The separate declaration before the references describes their use during manuscript preparation.

5. Reproducibility and replication protocol

The reproduction package consists of the frozen model configuration, the public-source retrieval procedures, the numerical supplementary tables, and the stepwise protocol below. The corresponding implementation and processed replay data are available from the author on reasonable request.

5.1. System development

Reconstruct the 18-feature ingestion schema, 72-step temporal window, physics-estimation paths, frozen channel weights, and locked second-training-run model described in Sections 3 and 4. Preserve the study dates, coordinates, thresholds, and blend weights when comparing results.

[5]The lunar terminator is the moving boundary between illuminated day and darkness, where rapid electrical and thermal changes may affect dust behavior.

[6]Regolith is the loose layer of fragmented rock and dust covering the lunar surface.

5.2. Qualitative framework evaluation

1. Apply the nine operational criteria and ordinal scale defined in Section 4.2.
2. Review the official documentation, open archives, and interfaces of the six single-purpose reference systems.
3. Score each system independently and do not combine the categories into a total rank.
4. Evaluate LIPAS with the same criteria to map structural and functional differences.

5.3. Fourteen-day quantitative studies

1. Use the equatorial reference, Apollo 11, and Nobile Rim coordinates.
2. Replay Window A (July 5 to July 18, 2026) and Window B (July 22 to August 4, 2026).
3. Extract the extended-dataset labels, including public drivers and documented physics/catalog anchors.
4. Generate forecasts with the locked second-training-run backend and frozen blend weights for all six hazards.
5. Reproduce the public-reference comparisons from archives, formulas, and engineering-model exports as described in Section 4.4.
6. Compare forecasts with the labels using absolute error, RMSE, and MAE. The full numerical values are provided in Supplementary Appendices C and D.

5.4. Mission-readiness simulation

1. Recreate the 7 h 37 min Apollo 17-length EV A timeline and the fixed caution/no-go thresholds.
2. Inject the controlled precursor, geomagnetic, and proton-flux changes at the times specified in Section 6.
3. Record warning-state changes for LIPAS, the GOES-only path, the replicated radiation path, and the combined NASA path.
4. Compare the number of products and steps required to obtain equivalent situational awareness.

5.5. Hazard ablation and sensitivity analysis

1. Run the complete model to establish baseline RMSE.
2. Replace each selected hazard input with its Window A mean while leaving the remaining data streams and weights fixed.
3. Repeat the defined two-channel replacements without retraining or retuning.
4. Record prediction changes relative to the full system as a sensitivity test.

6. Experiment-specific evaluations and findings

6.1. Experiment 1: qualitative comparison

6.1.1. Method

The qualitative comparison classifies each system across nine criteria: hazard coverage, data fusion, temporal forecasting, spatial coverage, real-time operations, robustness and stability, validation strength, usability and operational readiness, and openness and accessibility. The ordinal scale and classification rules are given in Section 4.2. Scores are categorical and are not combined into a numerical rank. Each system is classified according to its primary function: DONKI is an event-monitoring system; Diviner is a measurement archive; LDEX is a measurement system with limited spatiotemporal scope; Apollo PSE is a historical catalog; MEM 3 is a statistical engineering model; GOES-18 is a real-time measurement system adapted from an Earth-orbit context; and LIPAS is an integrated predictive system combining measurements, models, and machine learning.

6.1.2. Results and interpretation

The matrix shows a trade-off between specialized scientific detail and multi-hazard operational utility. Diviner and Apollo PSE, for example, have strong robustness and validation because they are foundational, peer-reviewed measurements with long scientific histories. Their specialized design, however, leads to low scores in hazard coverage, data fusion, and temporal forecasting, requiring planners to consult multiple products.

LIPAS receives Very High ratings for hazard coverage, data fusion, and usability because it integrates six environmental variables and provides a browser-based operational overview. Its High ratings for forecasting, spatial coverage, real-time operations, and openness reflect multi-day outputs for user-selected lunar locations, an automated pipeline, and public access. Moderate ratings for robustness and validation reflect external data dependencies and evaluation over limited, localized windows with substantial proxy targets. These classifications are based on published descriptions and available documentation; boundaries between categories require judgment. The matrix represents systems in their evaluated operational or research context, not their maximum theoretical capability.

The table shows where LIPAS adds value and where it does not. It scores well for fusion, unified multi-hazard support, and usability as a combined decision-support layer. It scores lower in specialized validation and hazard-specific accuracy where a dedicated agency model or measurement archive owns the problem. LIPAS is intended to complement those systems, not replace flight-qualified single-purpose models.

6.2. Experiment 2: prototype diagnostics and model redesign

6.2.1. Method

The preliminary studies evaluated LIPAS V1 in a daytime experiment from April 5 to April 19, 2026, at 0°N, 0°E. One study tracked all six hazards against replicated NASA and NOAA references; another examined short- versus long-horizon temperature behavior against a Diviner-style dayside relation. Their purpose was diagnostic rather than operational validation. They omitted secondary environmental factors, used only a subset of available agency data, and did not fully reproduce all mathematical components of the specialized models.

6.2.2. Results and interpretation

The preliminary comparisons used limited public-method replicas and do not establish performance for the current system. They did show that the client-side prototype handled temperature poorly when it relied on machine learning rather than known physical relationships. This motivated a physics-led temperature path and a backend for more consistent data handling and forecasting. The reported evaluation used the hybrid framework described in Sections 3 and 4. Full prototype data are provided in Supplementary Appendices A and B.

6.3. Experiment 3: locked-model multi-hazard evaluation

6.3.1. Method and validation scope

LIPAS V2 was evaluated over two independent 14-day windows using the replicated baselines described in Section 4.4. The same three sites were used in both windows, yielding 42 site-days per window.

The errors below quantify agreement with the channel-specific labels in Table 1. They are not six-hazard accuracy estimates against simultaneous lunar-surface ground truth. Lower error supports improved behavior relative to the stated replica or shared physics path within this controlled framework; it does not establish superiority over an internal NASA/NOAA operational product or a direct instrument measurement.

6.3.2. Results: Window A, predominantly lunar night

Window A covered July 5 to July 18, 2026. Local solar time at the study longitudes was dayside for approximately 19% of samples, so the interval is treated as the night regime. Against the replicated reference paths, hybrid solar-activity RMSE was 3.95 versus 4.66 (approximately 15% lower), and radiation RMSE was 0.102 versus 0.132 (approximately 22% lower). Temperature, moonquakes, and dust were ties because both predictions used the same physics path, although absolute site RMSE varied. The micrometeorite RMSE values were 0.011 for LIPAS and 0.109 for the no-shower MEM 3 comparison. This is a descriptive comparison of shower-term treatment, not evidence of general accuracy or performance against the full MEM framework.

6.3.3. Results: Window B, predominantly lunar day

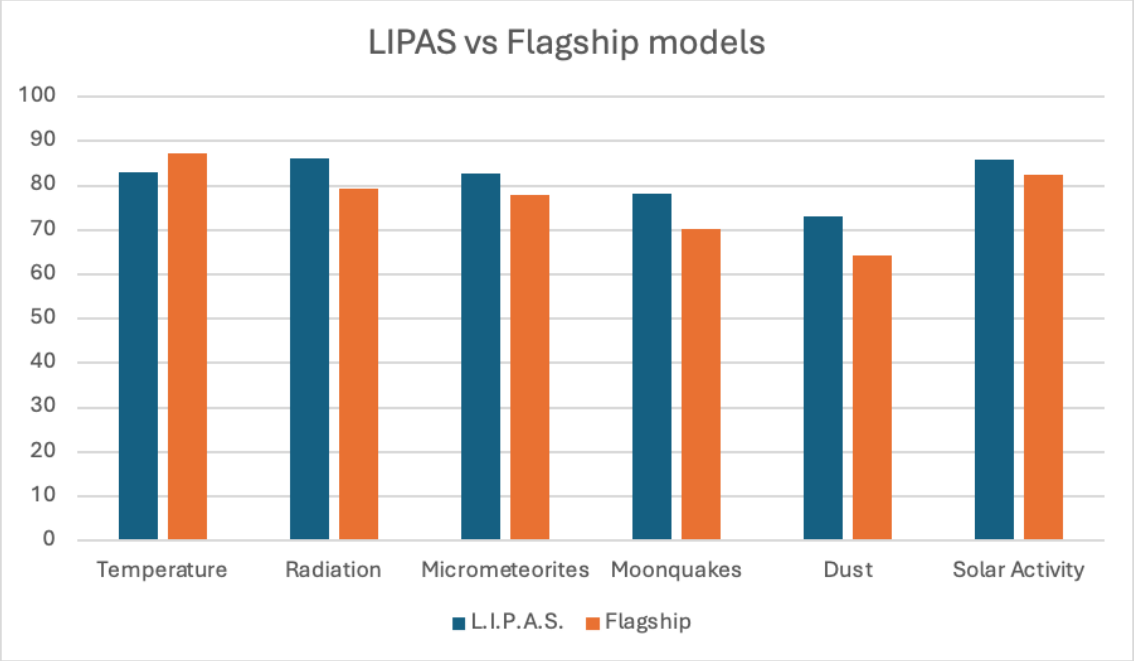
Window B covered July 22 to August 4, 2026. Approximately 95% of site-days were on the dayside. LIPAS used the MEM 3 seasonal envelope plus a shower-timing term, including the Perseid peak near day-of-year 216 (August 4), while the replicated micrometeorite comparison used the seasonal envelope without shower modulation.

Solar-activity RMSE was 4.99 versus 5.99, a modest hybrid improvement. Temperature, moonquakes, and dust remained ties on the shared physics paths. Micrometeorite comparison RMSE was 0.031 versus 0.309, but this is descriptive of the added shower-timing term against a seasonal baseline and does

Table 2: Qualitative comparison of LIPAS with selected agency models and archives. Criteria follow Section 4.2; ordinal categories are not summed. VH = Very High; H = High; M = Moderate; L = Low; VL = Very Low; N = None.

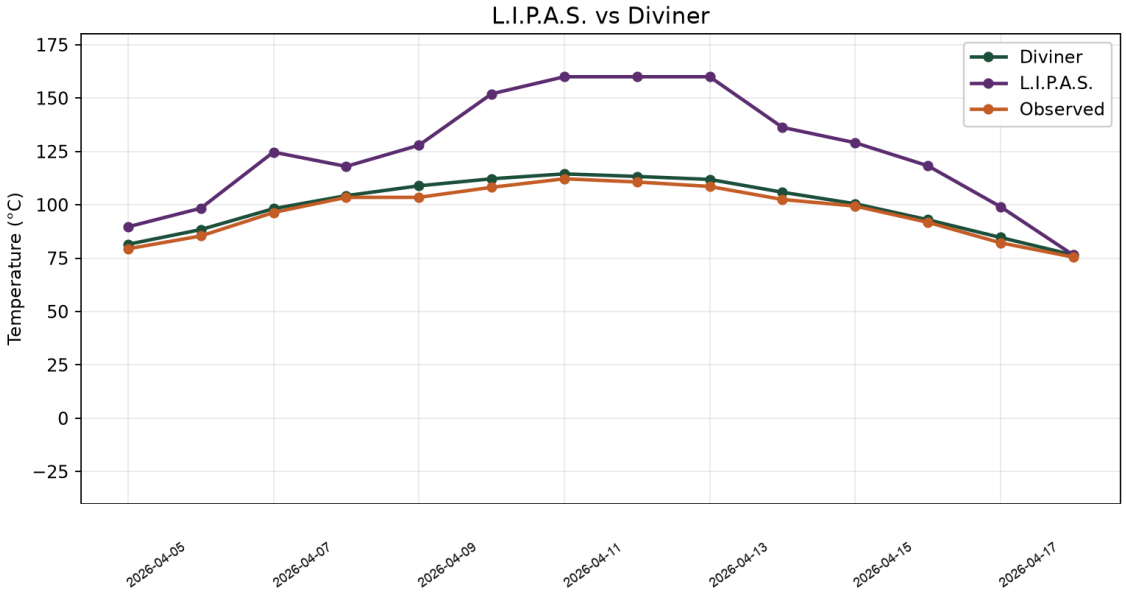
Model	Hazard	Fusion	Forecast	Spatial	Real-time	Robust.	Valid.	Usability	Open
LIPAS	VH	VH	H	H	H	M	M	VH	H
DONKI	L	L	L	M	H	H	H	M	H
Diviner	VL	L	L	VH	L	VH	VH	L	M
LDEX	VL	L	N	L	N	M	H	VL	M
Apollo PSE	VL	L	N	VL	N	H	VH	L	H
MEM 3	VL	L	M	H	L	H	H	L	M
GOES-18	L	M	M	M	VH	VH	VH	M	H

LIPAS criterion	Score	Basis for score
Hazard coverage	VH	All six study hazards appear in one system.
Data fusion	VH	Public space-weather feeds and lunar archives feed one hybrid pipeline.
Temporal forecasting	H	Multi-day forecasts are provided, but operational validation remains limited.
Spatial coverage	H	Users can request forecasts for lunar coordinates across the mapped surface.
Real-time operations	H	Automated external-feed ingestion supports timely updates, with upstream dependencies.
Robustness and stability	M	The system depends on public APIs and has limited long-window testing.
Validation strength	M	Structured comparisons are available, but many targets are proxies rather than surface measurements.
Usability and readiness	VH	A unified interactive dashboard combines hazard views and mission-planning tools.
Openness and accessibility	H	The browser interface and underlying source products are publicly accessible, but the full implementation is not an open operational service.

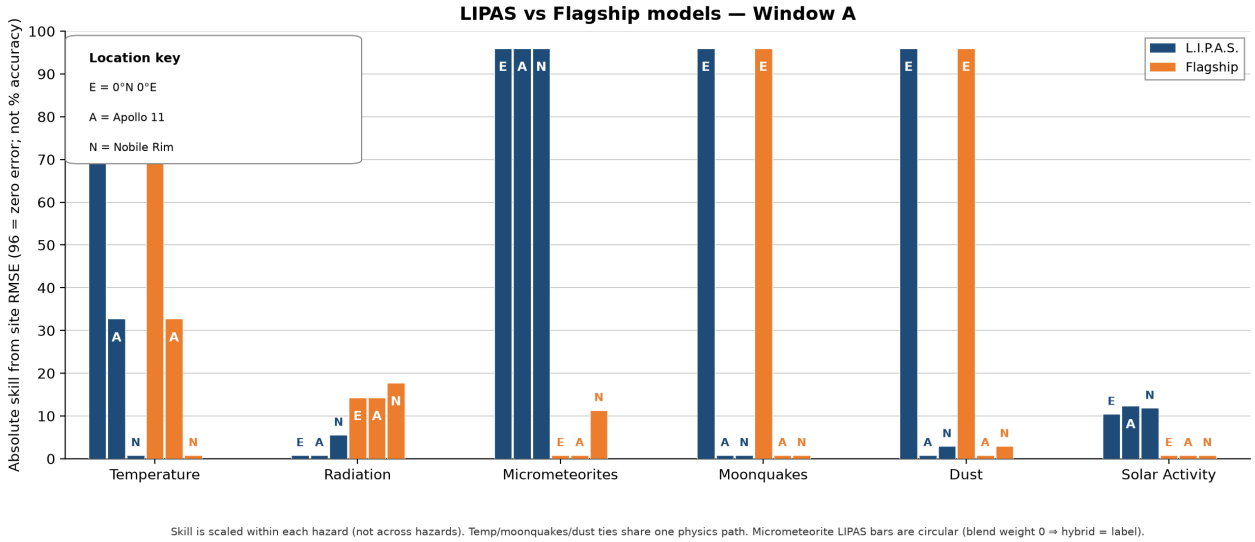


6.3.4. Interpretation

Across both windows, temperature, moonquakes, and dust match their shared physics paths by design (blend weight 0), so navy and orange skill bars coincide at each site. Site-to-site height differences within those hazards reflect different absolute RMSE, not model disagreement. Solar activity shows a modest hybrid RMSE reduction in both windows (4.10 versus 4.66 in Window A; 4.99 versus 5.99 in Window B). Radiation does not improve on the replica. Locked-model hybrid RMSE is higher in Window A (0.119 versus 0.102) and more clearly in quiet Window B (0.0188 versus 0.00266), where absolute doses are small, ML predictions are biased low, and the 0.05 blend still pulls away from the cleaner physics/replica path. Micrometeorite LIPAS skill is circular (hybrid equals the shower-augmented label) and is descriptive only. Bars were not height-normalized across hazards so these failures remain visible.



Figur n the meth h.

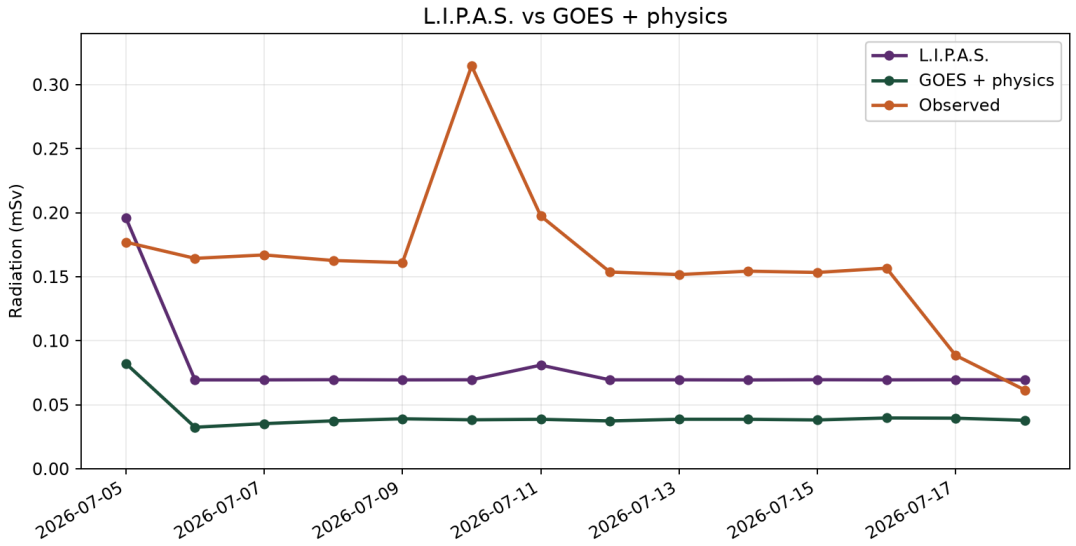


Figur Figure 6: Window A per-site absolute skill (July 5–18, 2026; 42 site-days). Skill = 96×max(0, 1−RMSE/scale) scaled within each hazard across sites — not is not perce height-normalized across hazards. Navy = LIPAS; orange = replica. Temp/moonquakes/dust ties share one physics path. Micrometeorite LIPAS bars are lidate lunar circular (blend weight 0). Radiation replica-ahead is a real locked-model result.

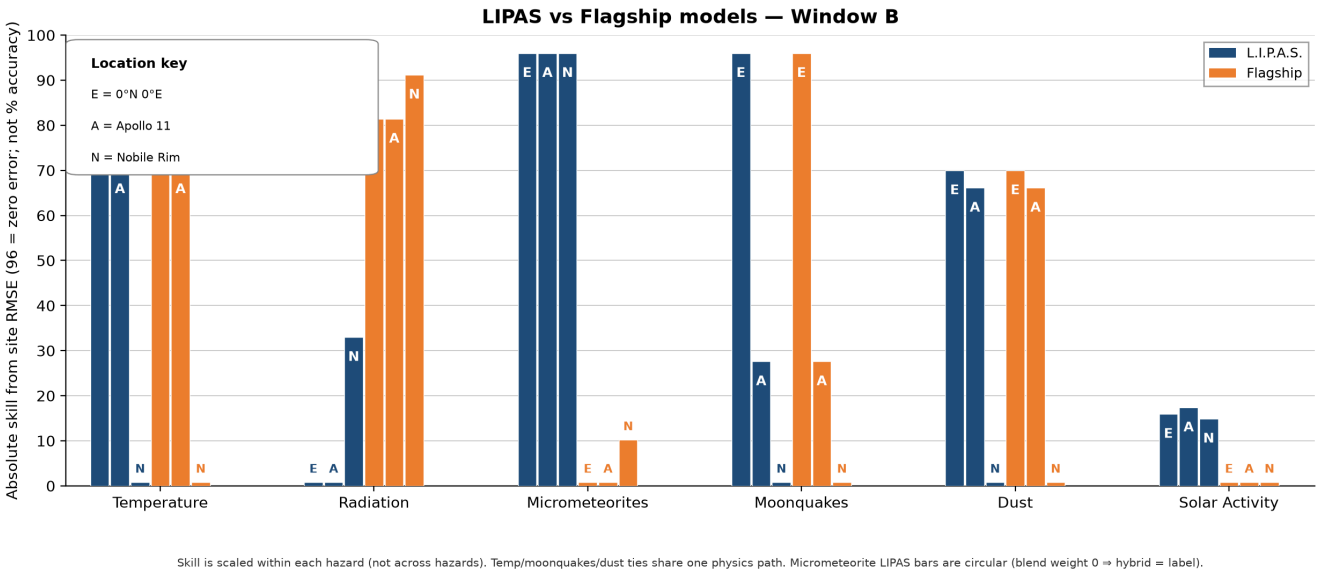
con
learn from. The small blend weight still pulls the result away from the lower-error physics/replica path. This is a quiet-regime radiation-channel limit, not evidence that LIPAS is broken. The micrometeorite comparison is limited to the stated shower term and seasonal MEM 3 baseline. All results remain conditional on the target provenance and replicated-reference definitions established in Section 4.

Table 3: Window A hybrid versus replicated-reference RMSE (July 5 to July 18, 2026; predominantly lunar night; 42 site-days).

Hazard	Hybrid	Replica	Outcome
Solar activity	4.10	4.66	Hybrid lower by ~12%
Radiation (mSv)	0.119	0.102	Replica lower (hybrid higher error)
Temperature (°C)	38.2	38.2	Tie on shared physics path
Moonquakes (day□¹)	1.23	1.23	Tie on shared physics path
Micrometeorite flux (×10□¹□)	0.0	0.624	Descriptive / circular (blend weight 0)
Dust (g cm□³)	0.526	0.526	Tie on shared physics path



FFigure 7: Window A radiation trace from the locked-model study, showing LIPAS, the GOES-plus-physics reference, and evaluation labels. This time series t accompanies the aggregate RMSE comparison in Table 3.



FFigure 8: Window B per-site absolute skill (July 22–August 4, 2026; 42 site-days). Same within-hazard skill as Figure 6 (no cross-hazard height normalization). Quiet o dayside radiation: hybrid RMSE 0.0188 versus replica 0.00266. Temp/moonquakes/dust are aggregate ties; micrometeorite LIPAS bars remain circular at skill 96.

Table 4: Window B hybrid versus replicated-reference RMSE (July 22 to August 4, 2026; predominantly lunar day; 42 site-days).

Hazard	Hybrid	Replica	Outcome
Solar activity	4.99	5.99	Modest hybrid improvement
Radiation (mSv)	0.0188	0.00266	Hybrid higher error in the quiet-dayside regime; ML-only 0.068, physics-only 0.016
Temperature (°C)	25.5	25.5	Tie on shared physics path
Moonquakes (day^{-1})	0.695	0.695	Tie on shared physics path
Micrometeorite flux ($\times 10^{-15}$)	0.0	0.701	Descriptive / circular (blend weight 0)
Dust (g cm^{-3})	0.169	0.169	Tie on shared physics path

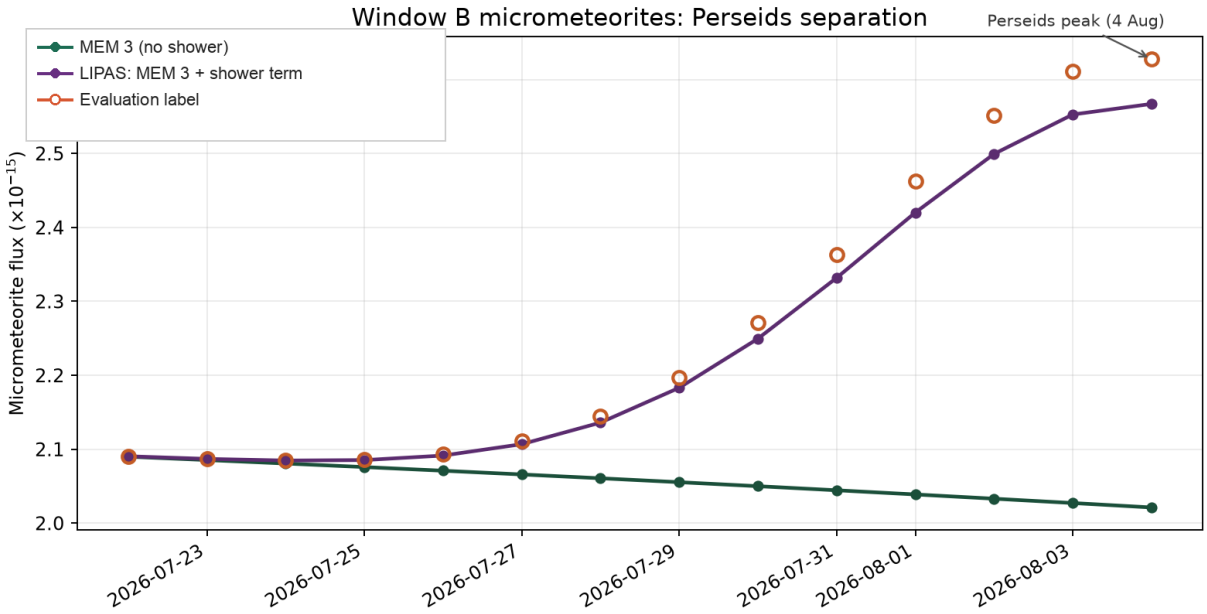


Figure 9: Window B study-site mean micrometeorite flux. The limited comparison shows MEM 3 with the added shower-timing term against MEM 3 without that term and the evaluation label. It does not validate the full MEM framework or independent lunar impact rates.

6.4. Experiment 4: operational mission simulation

6.4.1. Method

A controlled radiation-focused EV A simulation compared warning times from the unified dashboard with separate public products. The locked second-run model was evaluated at Apollo 11 over 7 h 37 min, the duration of Apollo 17 EVA 3. Quiet site drivers from July 12, 2026, were advanced in five-minute steps, and a storm scenario was injected so that all pathways faced identical conditions: the EV A began under quiet conditions at T+0; a solar/CME-style precursor began at T+40; rising Kp and southward IMF B_z began at T+55; and a GOES-like proton surge began at T+70 before peaking and decaying.

Identical radiation thresholds were applied to LIPAS and the NASA physics replica: caution at 0.095 mSv h^{-1} and no-go at 0.15 mSv h^{-1}. A GOES-only path used thresholds on proton flux. A combined NASA path selected the worse of the physics-radiation status and GOES proton status, approximating an operator monitoring both products.

6.4.2. Results and interpretation

LIPAS first issued caution at T+45, five minutes after the injected precursor and 25 minutes before the proton surge. Its use of upstream solar and geomagnetic drivers explains the earlier response relative to the GOES-only path; the result does not imply that LIPAS detects events GOES cannot detect. LIPAS reached no-go at T+85. GOES alone reached caution at T+80 and no-go at T+105. The NASA physics replica reached caution at T+55 and no-go at T+70, so the combined NASA path produced caution at T+55 and no-go at T+70.

The simulation was intentionally radiation-focused and is not evidence that LIPAS is superior for every hazard. NASA architecture studies emphasize real-time space-weather situational awareness and operational response time for human deep-space missions (Minow et al., 2020). The combined-NASA path should be read as an idealized simultaneous-monitoring condition; this study did not measure an actual EV A operator workflow. LIPAS warned earlier than the GOES-only path, but the combined NASA path produced the earlier hard no-go in this controlled injection.

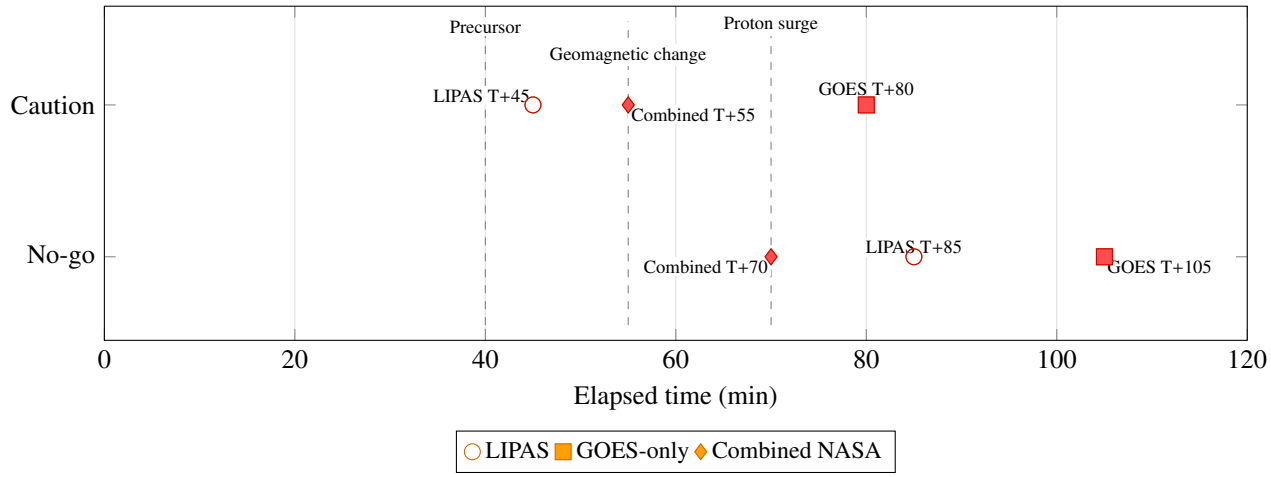


Figure 10: Mission-simulation decision timeline. Marker shape identifies the information pathway; amber markers denote first caution and red markers denote first no-go. Dashed vertical lines show the injected precursor, geomagnetic change, and proton surge.

6.5. Experiment 5: ablation of cross-hazard inputs

6.5.1. Method

The ablation experiment was a sensitivity test of selected hazard channels and driver inputs. Window A site-days were replayed with one input replaced by its Window A mean; two-input replacements were also tested. The model was not retuned. Performance was measured using solar-activity and radiation RMSE.

6.5.2. Results and interpretation

Setting solar activity to its mean was associated with higher radiation RMSE, from 0.1020 to 0.1050 (approximately 2.9%). Setting radiation itself to its mean produced only a small change, while the remaining individual hazards had negligible effects. Setting both solar activity and radiation to their means increased RMSE to 0.1068, approximately 4.7% worse than the full model. The Window B quiet-dayside result also shows why radiation-

7. Cross-experiment discussion

RQ1 (development). A channel-specific hybrid approach was more appropriate than applying machine learning uniformly. Physics-tied channels stay on shared ns vers add information, especially solar activity, not radiation.

RQ2 (usability). The unified dashboard improved how information is presented for planning, but this study did not show that LIPAS is safer than existing agency pathways.

The first research question concerned how LIPAS should be developed and what could be learned from the original prototype. Machine learning was not equally useful for all six hazards. Treating every hazard primarily as a machine-learning problem produced unnecessary variation in channels already constrained by established physical relationships. The current hazard-specific hybrid architecture therefore leaves physics dominant where it provides a strong foundation and gives machine learning greater influence where additional predictive information is plausible. Window B adds a boundary condition: with little storm signal on quiet dayside samples, the neural correction has little useful work and the small radiation blend mostly adds error. A channel-specific hybrid approach was more appropriate than applying machine learning uniformly.

The second research question asked whether integrating environmental hazards could improve operational usability. The qualitative comparison and controlled simulation provide a partial answer. NASA and NOAA models and archives remain more

RQ3 (accura ity - n show lower solar RMSE in both Window A and Window B show lower solar RMSE in both Window A and Window B veat solute doses are small (0.0188 versus 0.00266). Temperature, moonquakes, and d Micrometeorite skill reflects the shower-timing term only.

Mission simulation. LIPAS issued caution earlier than the GOES-only path by using upstream drivers; the combined NASA path still produced the earlier hard no-go. Treat LIPAS as consultative decision support alongside agency products.

Ablation. Setting solar input to its Window A mean lowered solar RMSE and raised radiation RMSE in that quiet interval. This is sensitivity, not evidence that live solar data are generally harmful.

Table 5: Radiation-channel sensitivity results from input ablation. Changes are full-model RMSE minus configuration RMSE.

Configuration	RMSE	Full minus case	Relative change
Full model	0.1020	0.0000	0.0%
Set solar to mean	0.1050	−0.0029	−2.9%
Set radiation to mean	0.1029	−0.0009	−0.9%
Set temperature to mean	0.1020	∼ 0	∼ 0%
Set moonquakes to mean	0.1020	∼ 0	∼ 0%
Set micrometeorites to mean	0.1020	∼ 0	∼ 0%
Set dust to mean	0.1020	∼ 0	∼ 0%
Set solar and radiation to mean	0.1068	−0.0048	−4.7%

Table 6: Solar-activity sensitivity results from input ablation. Changes are full-model RMSE minus configuration RMSE.

Configuration	RMSE	Full minus case	Relative change
Full model	3.899	0.000	0.0%
Set micrometeorites to mean	3.984	−0.085	−2.2%
Set radiation to mean	3.933	−0.034	−0.9%
Set moonquakes to mean	3.931	−0.032	−0.8%
Set temperature to mean	3.916	−0.018	−0.5%
Set dust to mean	3.909	−0.011	−0.3%
Set solar to mean	3.290	+0.608	+15.6% (lower RMSE)
Set moonquakes and dust to mean	3.960	−0.061	−1.6%

for the hybrid and for the replica; ML-only RMSE was about 0.068 and physics-only RMSE was about 0.016. The hybrid beats raw ML but has higher error than the replica. This is a quiet-regime radiation-channel limit, not evidence that LIPAS is broken. Temperature, moonquakes, and dust remained aligned with their physics references. The micrometeorite comparison was limited to adding shower timing against a seasonal MEM 3 baseline and does not establish performance against the full framework. The answer is therefore selective: the holdout result and solar comparisons are encouraging, but the proxy labels and replicated baselines limit broader conclusions.

The absence of improvement in temperature, moonquakes, and dust is a consequence of design rather than necessarily a weakness. Those channels were deliberately physics-led. Matching their physics paths indicates that the hybrid did not introduce unnecessary neural corrections. The uncertainty-guided weight in Eq. (3) is central to this behavior. LIPAS does not simply average neural and physical predictions; the neural contribution changes with estimated uncertainty and environmental conditions. When the learned prediction becomes less certain, the physics path becomes more influential.

The clearest evidence for additional learned information came from the modest solar-activity reductions rather than any radiation will over the replica.

rather than any radiation will over the replica.

The sensitivity test found higher radiation RMSE when the solar input was replaced by its Window A mean. This is consistent with a contribution from solar context in this test, although Window B shows that radiation performance depends on regime: its quiet dayside conditions leave little storm signal, and the small blend mostly adds error. These findings require caution because the evaluation targets partly comprise estimated or physics-based values rather than continuous co-located lunar-surface measurements. The comparisons use replicated public baselines, not operational NASA or NOAA software. They demonstrate behavior within the experimental framework, not that LIPAS measures lunar radiation more accurately than a spacecraft instrument.

The mission simulation illustrates why predictive error and operational usefulness are not identical. In the controlled scenario, LIPAS produced caution earlier than the GOES-only path because it incorporated injected precursor signals before the proton-flux threshold was crossed. The combined NASA path nevertheless produced the earlier no-go. Different warning times follow from the signals monitored and how those signals are mapped to thresholds. The simulation does not show detection of a real storm or an event unavailable to NASA products.

The micrometeorite results demonstrate a limited comparison of shower timing. During the Perseid period, LIPAS included an explicit shower term while the selected MEM 3 comparison used only a seasonal background. The predictions diverged as the shower approached its peak. This contrast does not independently validate lunar impact rates or establish performance against the full MEM framework.

The ablation results further demonstrate sensitivity to selected inputs, not causal cross-hazard effects. Replacing the solar-activity input with its Window A mean increased radiation error, while other input replacements had much smaller effects. Replacing the live solar input with its Window A mean reduced solar error, suggesting that short-term input variation, quiet-condition weighting, or the construction of the evaluation labels favored a smoother signal during that window. This sensitivity result does not show that live solar data are generally harmful; stronger-activity testing is needed.

Combining physics and machine learning is not itself new (Karpatne et al., 2017; Willard et al., 2022; Camporeale, 2019). The contribution of LIPAS is the application of hazard-specific weighting, uncertainty information, and cross-hazard inputs within a unified lunar decision-support system. The results provide preliminary support while also showing that the architecture is not a replacement for specialized scientific models or a fully validated operational forecast system.

Several limitations constrain the conclusions. Quantitative targets are constructed from public observations, physical formulations, historical catalogs, and proxies rather than continuous co-located lunar-surface measurements. The results therefore describe comparative behavior within the evaluation framework, not direct accuracy against complete lunar-surface truth. Except for the available MEM 3 engineering-model export, the agency comparisons are independent reconstructions of documented public methods rather than executions of proprietary operational systems. Their value is methodological comparability under shared inputs, dates, and locations; they should not be interpreted as certified reproductions of every internal agency component. LIPAS's strongest demonstrated value is consequently as a common layer in which established physics, live space-weather information, cross-hazard relationships, and uncertainty can be considered together.

This limitation is also testable rather than permanent. Artemis-era surface instruments, commercial landers, and future lunar geophysical networks can progressively replace proxy labels with time-matched measurements. The target-provenance hierarchy in Table 1 provides a direct upgrade path: each proxy can be superseded without changing the overall evaluation design, enabling later studies to distinguish predictive performance from agreement with an assumed physical path.

8. Future work

Further work is required before LIPAS can be regarded as an operationally trusted forecasting tool. The main limitation is the absence of continuous surface measurements across all six channels. Future work should build on the completed night/day evaluations, mission simulation, and ablation results rather than repeat the same experiments.

8.1. Improving physics-tied channels

Temperature, moonquakes, and dust matched their reference paths as intended. The micrometeorite comparison tested a stated shower-timing term against a seasonal MEM 3 baseline, rather than the full framework. Future work should strengthen shower timing and MEM 3 integration rather than treating micrometeorites as a purely machine-learning target.

8.2. Understanding the solar ablation result

In the sensitivity test, replacing the solar input with its Window A mean was associated with lower solar-activity RMSE and higher radiation RMSE. These results are limited to the relatively quiet Window A interval. The tests measure sensitivity to selected inputs, not causal cross-hazard effects, and the lower solar error does not establish that live solar data are generally harmful. Future work should determine whether this behavior is caused by input noise, target smoothness, or regime-dependent weighting.

8.3. Expanding solar and radiation models

Solar activity showed modestly lower error in both windows. Radiation error was also lower in both ow B reported under a quiet-window caveat (0.0188 versus 0.0026). The reported radiation gain should therefor ther than as a storm-period result.

Future work should evaluate the radiation channel across a wider range of activity, with denser OMNI, DONKI, and SWPC sampling, improved treatment of proton-flux events, calibrated uncertainty estimates, and validation against additional historical space-weather events. Differences between LIPAS and combined-NASA warning paths should be examined transparently rather than resolved by retrospective threshold tuning.

8.4. Stronger validation with measured data

As observations from Artemis missions, landers, rovers, and orbital instruments become available, estimated labels should be replaced with direct measurements wherever possible. The highest-value additions are co-located time series measuring radiation, thermal state, dust, impacts, and seismic activity at the same site. Where measured values remain unavailable, proxies should remain explicitly classified by provenance. Collaboration with NASA, NOAA, ESA, JAXA, commercial lunar payload teams, and university groups could improve access to datasets, operational expertise, and mission-relevant performance criteria. LIPAS should continue to be presented as decision support under incomplete observations, not as a replacement for scientific instruments or engineering models.

8.5. Extending beyond the lunar surface

Because the framework already uses heliophysics data, cislunar space[7] and Earth orbit are logical extensions, with Mars as a possible longer-term application (Cucinotta et al., 2010; Hassler et al., 2014; Zeitlin et al., 2013). These environments offer more observational data and therefore stronger opportunities to validate the hybrid methodology. The channel-specific weighting architecture could be reused without a complete redesign.

[7] Cislunar space is the region between Earth and the Moon, including lunar orbit and the trajectories connecting the two bodies.

8.6. Operational development

Future dashboard development should improve uncertainty visualization, mission-specific thresholds, and reports aligned with operational checklists. Interface improvements are useful only when they communicate the underlying scientific results accurately and transparently.

8.7. Future-work summary

The immediate priorities are to strengthen physics-tied channels, explain the solar ablation behavior, and further validate solar and radiation models. Over the medium term, new measurements and collaboration with space agencies should improve confidence. Longer term, the same framework could support cislunar and eventually Mars operations. Throughout development, LIPAS should remain a hybrid decision-support platform for incomplete observations rather than a universal replacement for specialized models or flight-qualified systems.

9. Conclusion

This research shows how machine learning can be integrated into a unified, physics-based lunar-forecasting system while retaining the physical foundations of established environmental models. The development process answered the first research question by showing that a hazard-specific hybrid approach was appropriate for the current model, rather than applying machine learning uniformly across all hazards. This balance may change as machine-learning methods improve. The operational evaluation answered the second by showing that integrating hazards and data sources can improve how information is presented during mission planning, but it did not establish that LIPAS is safer than existing approaches. A unified system can also concentrate risk if its data or service fails. LIPAS is best considered alongside current NASA and NOAA systems as a consultative decision-support tool, subject to validation with real mission workflows and measurements.

The quantitative evaluation answered the third selectively. Chronological holdout RMSE was about 4.97 versus about 13.66 for persistence, and solar activity improved modestly in both windows. Radiation did not: hybrid RMSE exceeded the replica in Window A (0.119 versus 0.102) and in quiet Window B (0.0188 versus 0.00266). Temperature, moonquakes, and dust remained physics ties. Micrometeorite skill is circular under blend weight 0 and remains shower-term descriptive only. These studies therefore teach where the hybrid helps and where it still fails.

The central result is that a carefully constrained hybrid architecture can add predictive capability where machine learning is useful without replacing physics-based methods where they are more appropriate. LIPAS's principal contribution is therefore not the replacement of existing lunar models, but their integration into a single forecasting and decision-support framework. The system combines established physical models, public space-weather data, learned inference, and cross-hazard context as a step toward operational multi-hazard forecasting.

The findings must be read with those limits in mind: proxy labels, quiet-regime radiation shortfalls, and circular micrometeorite scoring. LIPAS should function alongside NASA and NOAA products as consultative decision support, not as a replacement. The studies show how much remains to develop—radiation under quiet and storm regimes, stronger validation with surface measurements, and clearer operational workflows. This work does not end here; it is a first constrained step that the next iterations must extend.

Acknowledgements

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CRediT author statement

Luka Takki: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing, original draft, and Writing, review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The evaluation uses public NASA, NOAA, OMNI, Diviner, CRaTER, LADEE/LDEX, Apollo, MEM 3, and meteor-shower sources listed in the references. Full numerical values used in the

reported experiments are reproduced in Supplementary Appendices A to E. Code and locked model files: https://github.com/Luka-bot-afk/LIPAS. Daily tables are in papers/appendix_numerical_tables/.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, I used Cursor and Google Gemini to review, and to help modify, selected parts of the code and the manuscript. These tools did not write the paper. I checked every suggested change, made the final edits myself, and take full responsibility for the content of the published article. Their role was limited to review and selective modification of existing material; they were not used to draft or author the manuscript.

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Appendix A. Numerical data tables

Go to:
https://github.com/Luka-bot-afk/LIPAS/tree/main/papers/appendix_numerical_tables

Open the file LIPAS_appendix_numerical_tables_A_B_D_E.pdf.

That file has the daily rows for the prototype April study and for Window A and Window B.

For the aggregate RMSE numbers in Tables 3 and 4, also open reported_window_rmse.md in the same folder.

Repository: https://github.com/Luka-bot-afk/LIPAS

Appendix B. Qualitative comparison framework

Go to:
https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/LIPAS_qualitative_comparison_framework.md

Open LIPAS_qualitative_comparison_framework.md.

Use those rules to score DONKI, Diviner, LDEX, Apollo PSE, MEM 3, GOES-18, and LIPAS on the nine criteria (None to Very High; written as N, VL, L, M, H, VH in Table 2).

Do not turn the labels into numbers and do not sum them. Compare your scores with Table 2.

Appendix C. Quantitative comparison framework

Go to:
https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/LIPAS_appendix_numerical_tables_A_B_D_E.pdf

Open LIPAS_appendix_numerical_tables_A_B_D_E.pdf for the daily values behind Tables 3–6.

What to compare
• Sites: 0°N 0°E, Apollo 11, Nobile Rim
• Window A: 5–18 July 2026
• Window B: 22 July–4 August 2026
• Hazards: solar activity, radiation, temperature, moonquakes, micrometeorites, dust

How to measure error
• Use RMSE of hybrid vs replica against the study labels
• Skill on the charts: 96 × max(0, 1 − RMSE / scale); 96 = zero error
• Site codes: E = 0°N 0°E, A = Apollo 11, N = Nobile Rim

For the ablation tables (Tables 5–6), replace one driver with its Window A mean and check how RMSE changes.

Commands to regenerate the runs are in Appendix E.

Appendix D. AI disclosure

I used Cursor and Google Gemini to review, and to help modify, selected parts of the code and the manuscript. These tools did not write the paper. I checked every suggested change, made the final edits myself, and I am responsible for the paper and the results.

Appendix E. Code repository and reproduction

Go to: https://github.com/Luka-bot-afk/LIPAS

Setup
1. git clone https://github.com/Luka-bot-afk/LIPAS.git && cd LIPAS
2. python3.13 -m venv .venv313 && source .venv313/bin/activate
3. pip install -r requirements.txt
4. Leave saved_model/ unchanged

Phase 1 — Table 2 (qualitative)
1. Go to https://github.com/Luka-bot-afk/LIPAS/blob/main/papers/appendix_numerical_tables/LIPAS_qualitative_comparison_framework.md
2. Score DONKI, Diviner, LDEX, Apollo PSE, MEM 3, GOES-18, and LIPAS
3. Compare with Table 2 (do not sum scores)

Phase 2 — Window A (Table 3, Figure 6)
Go to: https://github.com/Luka-bot-afk/LIPAS/blob/main/scripts/run_14day_flagship_study.py
Run:
python scripts/run_14day_flagship_study.py --start 2026-07-05 --end 2026-07-18 --window-label A --out-dir logs/study_14day_window_A_night --model-dir saved_model
Then open logs/study_14day_window_A_night/summary.json and skill_by_site.png

Phase 3 — Window B (Table 4, Figure 8)
Run:
python scripts/run_14day_flagship_study.py --start 2026-07-22 --end 2026-08-04 --window-label B --out-dir logs/study_14day_window_B_day --model-dir saved_model
Then open logs/study_14day_window_B_day/summary.json and skill_by_site.png
For daily rows, go to papers/appendix_numerical_tables/ and open LIPAS_appendix_numerical_tables_A_B_D_E.pdf

Phase 4 — Mission simulation (Section 6.4, Figure 10)
Go to: https://github.com/Luka-bot-afk/LIPAS/blob/main/scripts/run_mission_sim_radiation_eva.py
Run: python scripts/run_mission_sim_radiation_eva.py
Compare caution / no-go times with Section 6.4

Phase 5 — Ablation (Tables 5–6)
Go to: https://github.com/Luka-bot-afk/LIPAS/blob/main/scripts/run_hazard_ablation_study.py
Run: python scripts/run_hazard_ablation_study.py
Compare RMSE changes with Tables 5 and 6

Do not change saved_model weights, blend weights, site coordinates, or window dates if you want a paper-faithful run.